

## Week 06: Basic Graph Theory

### 1 Exercises

**Exercise 1.** Does there exist a simple undirected graph with 6 vertices, 10 edges, where 5 of its vertices have degrees 1, 3, 3, 3, 5?

**Exercise 2.** Construct a regular graph for the following cases:

- a) 2025 vertices where every vertex has degree 10.
- b) 2026 vertices where every vertex has degree 11.

**Exercise 3.** Consider a  $4 \times 4$  chessboard. Two squares are "linked" if a knight can move from one to the other.

- a) Construct graph  $G_4$ . Find the degree of each vertex.
- b) Calculate the total ways to place 2 identical knights so they attack each other.
- c) Calculate the total ways to place 3 identical knights such that one knight attacks the other two.

**Exercise 4.** In a round-robin tournament with  $n$  teams ( $n \geq 4$ ),  $(n + 1)$  matches have been played. Prove at least one team has played at least 3 matches.

**Exercise 5.** Let  $G = (V, E)$  and  $A \subset V$ . Let  $k$  be the number of odd-degree vertices in  $A$ . Let  $m$  be the number of edges with one endpoint in  $A$  and one in  $V \setminus A$ . Prove  $(k - m)$  is an even integer.

**Exercise 6.** Find the number of vertices  $n$  for:

- a) 12 edges and all vertices have degree 3.
- b) 10 edges and all vertices have degree 4.
- c) 21 edges, three vertices of degree 4, others degree 3.

**Exercise 7.** A graph  $G$  has 19 edges and one vertex with degree  $\geq 3$ . What is the maximum number of vertices?

**Exercise 8.** Prove  $n - p \leq m \leq \frac{1}{2}(n - p)(n - p + 1)$  for a graph with  $p$  components. Show if  $2m > (n - 1)(n - 2)$ ,  $G$  is connected.

**Exercise 9.** Do simple graphs with the following degree sequences exist?

- a) 5, 3, 3, 2, 1, 0
- b) 6, 5, 3, 3, 2, 1
- c) 3, 3, 2, 2, 2, 2

d) 5, 3, 3, 3, 3, 3

e) 1, 1, 1, 1, 1, 1